CAMPYTEQ ENTERPRISE SYSTEM SPECIFICATION

Unified Higher-Ed ERP & AI Intelligence • Multi-Role Governance & Formal Data Flow Diagrams

1. System Architecture & Tech Boundaries

CampyTeq operates strictly on a clean, production-grade architecture: **React (Vite) Single Page Application** with Tailwind CSS, **Django 5.x REST Framework** backend in Python 3.12, **PostgreSQL** relational database, and native **Python AI & Computer Vision** analytics. Strict multi-tenant database isolation ensures complete security and zero unnecessary third-party brokers (Docker, Redis, Next.js, and mobile packages are eliminated).

2. Account Capabilities & Authorization Matrix

Role-based access control enforces granular permissions across all institutional stakeholders:

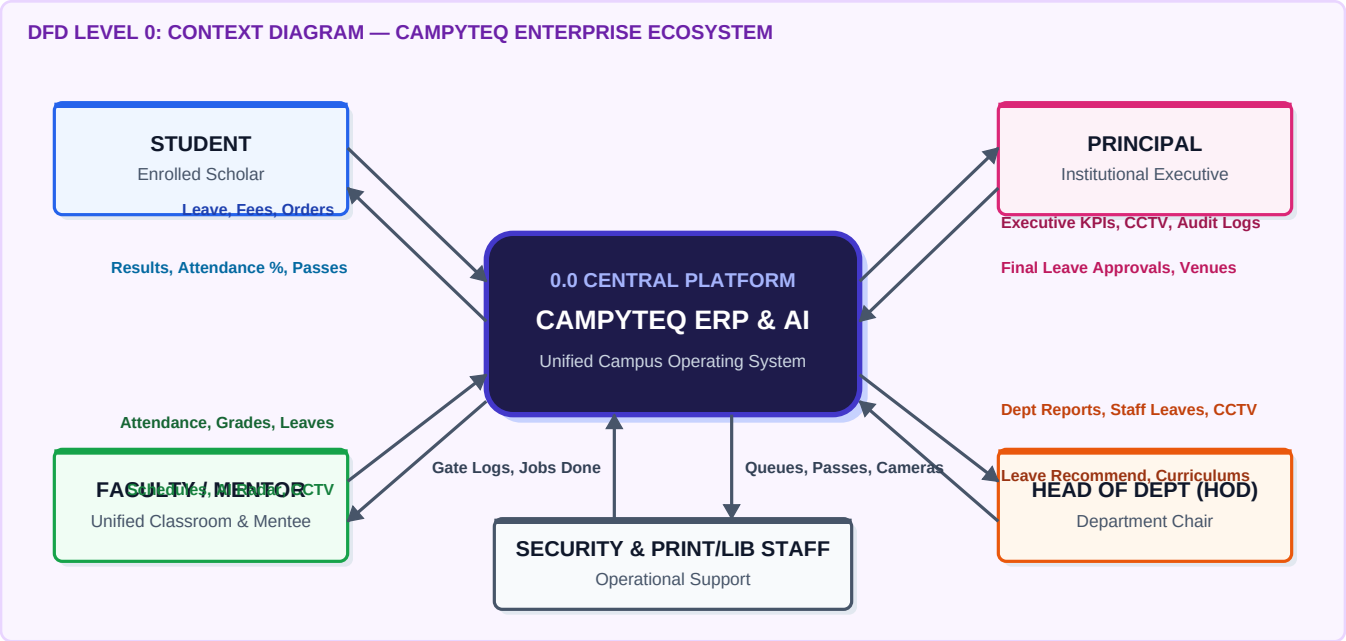
Role / Account	What They Can See (Visibility)	What They Can Do (Actions & Approvals)
SUPERADMIN	• All Colleges / Multi-tenant directories • Global storage & DB metrics • Platform-wide Audit Logs	• Provision new college tenants • Configure institutional subscription plans • Reset institution root admin credentials
PRINCIPAL	• Executive KPI Dashboard (Enrollment, Fees) • All 6 Departments (Students, Faculty, Staff) • **Authorized CCTV Cameras (All Zones)** • AI Radar Retention Risk Heatmaps • Institution Payroll & Fee Reports *(Hidden: Print Shop & Apply Leave)*	• **Final approval for Faculty & HOD leaves** • Approve venue & seminar hall bookings • Publish institution-wide announcements • Sign off staff payroll registers
HEAD OF DEPT (HOD)	• Departmental Student & Faculty rosters • Department Course Curriculums & Timetables • **Departmental CCTV Cameras** • Department AI Radar Risk metrics • Department Faculty Leave Requests	• **Review Faculty leaves & recommend to Principal** • Assign subjects, timetables, and labs to Faculty • Escalate student attendance/risk issues • Approve student internal assessment results
FACULTY / MENTOR (Unified)	• Assigned Classes, Timetables, & Syllabus • Student Rosters & Attendance registers • Mentee List (Mentorship Dashboard) • Mentee AI Risk radar & performance trends • **Authorized CCTV (Labs & Classrooms)** • Library Catalog & Campus Print Queue	• Mark daily lecture attendance • Create, publish, & grade assignments & exams • **Review & Approve/Reject assigned Mentee leaves** • Record 1-on-1 counseling / mentorship notes • **Apply for personal leaves (auto-routed to HOD)** • Book lab facilities & submit print jobs
STUDENT	• Academic Dashboard & Timetable • Attendance % breakdown & warnings • Course Assignments & due dates • Exam schedules & Grade cards • Outstanding Fee Invoices & Receipts • Library Catalog, loans & reservations	• **Apply for leaves (auto-routed to Mentor)** • Submit assignment files & track scores • Pay college fees online & download receipts • Submit document print requests to Print Shop • Reserve library books & check fine balances
SECURITY OFFICER	• **Authorized CCTV Feeds (Gates, Perimeter)** • Visitor log & Vehicle entry directory • Student gate-pass approval status	• Scan student RFID/QR gate passes on entry/exit • Log visitor identity, host person, and purpose • Raise security emergency alerts to Principal
ACCOUNTANT / FINANCE	• Fee Structures & Payment Schedules • Student Collections, Dues, & Defaulters • Salary & Payroll registers for all staff	• Generate fee invoices & concessions • Process payment reconciliations & refunds • Disburse monthly staff salary slips
PRINT SHOP STAFF	• Real-time Print Order Queue & specs • Uploaded PDF document previews • Daily print shop revenue & paper stock	• Accept incoming print jobs & execute hardware prints • Update status: Pending → Ready for Pickup → Done • Collect cash/counter payments if unpaid

LIBRARY STAFF	• Library Catalog (Books, ISBN, copies) • Active Issue/Return circulation register • Overdue loans list & accumulated fines	• Add/update book copies, racks, & barcode tags • Issue books to students/faculty via barcode scan • Receive returned books & clear fines upon receipt

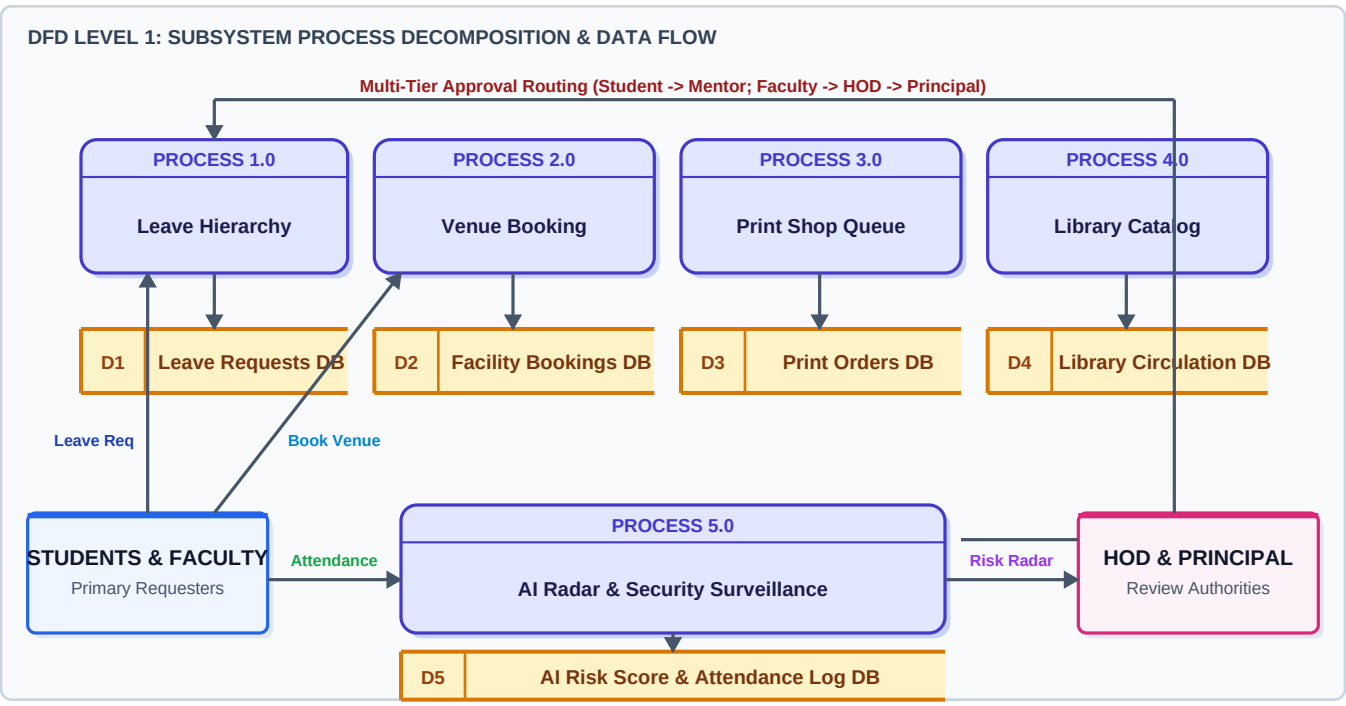
3. Data Flow Diagrams (DFD)

The following vector diagrams represent the formal Data Flow Architecture across all operational subsystems.

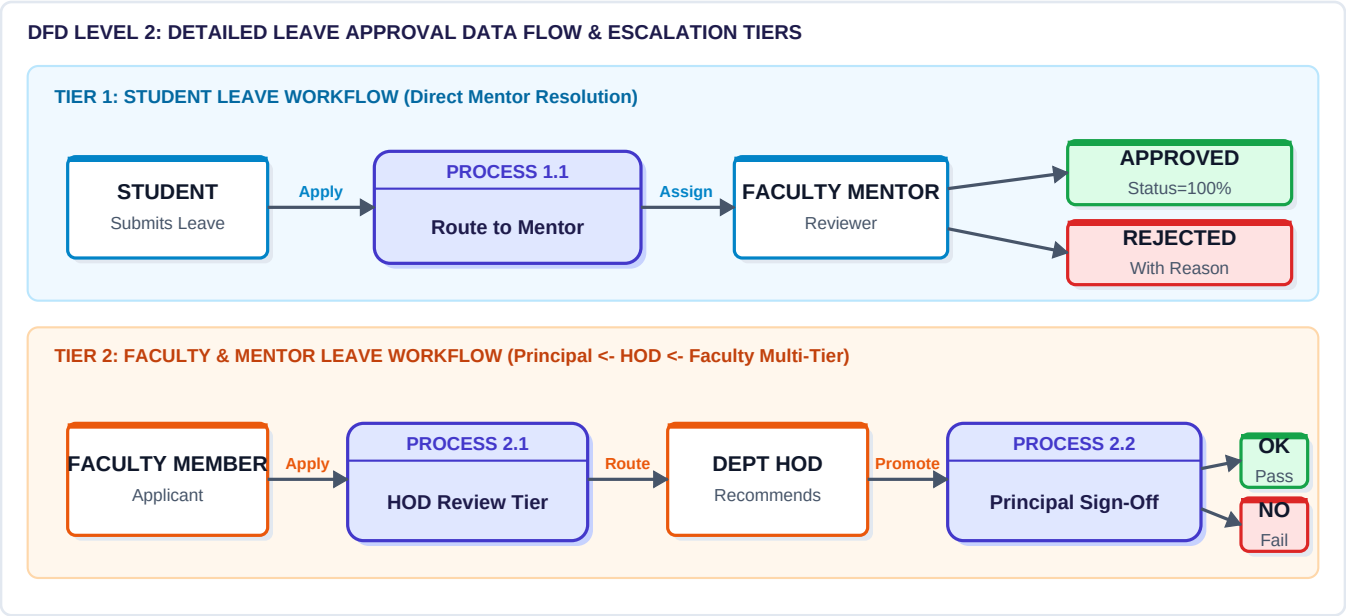
A. DFD Level 0: Global Context Diagram



B. DFD Level 1: Subsystem Process Decomposition



C. DFD Level 2: Multi-Tier Leave Request Workflow



4. End-to-End Request Lifecycles

1. Multi-Tier Leave Approval Hierarchy:

- **Student Leave Sub-flow:** The student submits dates, leave type, and reason. The backend automatically determines the student's assigned Mentor (Faculty profile) and sets routed_tier='MENTOR'. The mentor reviews and issues an immediate decision (APPROVED or REJECTED with feedback).
- **Staff Leave Sub-flow (Principal ← HOD ← Faculty):** When a faculty member applies for leave, it is routed to their Department HOD (routed_tier='HOD'). Upon HOD recommendation, the system promotes the request to routed_tier='PRINCIPAL'. The Principal grants final institutional sign-off, notifying both the applicant and HOD.

2. Campus Print Shop Service:

- Students and Faculty upload PDF documents, choose print properties (Color / Black & White, Single / Duplex, Copies).
- The system dynamically calculates tariffs (e.g. ■2/page BW, ■5/page Color). The order is queued for the Print Shop Staff.
- Staff marks job PROCESSING → READY_FOR_PICKUP (push alert dispatched to user) → user collects hardcopy at counter → COMPLETED.

3. Venue & Facility Booking:

- Faculty, Mentors, and HODs browse seminar halls, auditoriums, and computer labs.
- Backend algorithm checks for zero schedule collisions across time intervals. The reservation is submitted for Principal / Admin sign-off.
- Upon approval, the facility calendar is locked and security/maintenance teams are notified.

4. Library Circulation & Fines:

- Users search the catalog and place 24-hour holds on available books.
- At the circulation desk, staff scans student barcode and book barcode, issuing the book for 14 days.
- If overdue, daily fine accumulation updates automatically until returned and cleared.

5. AI Radar Risk Intelligence & CCTV Surveillance:

- **CCTV Camera Authorization:** Live feeds are strictly restricted to **Principal, Security, HOD, and Faculty**.
- **AI Radar Computation:** Real-time Python AI algorithms aggregate lecture attendance velocity, assignment completion, and test marks to classify students into **HIGH RISK** (>75% dropout probability), **MEDIUM RISK**, or **LOW RISK**.
- Retention intervention notices are dispatched immediately to the assigned Faculty Mentor and HOD for 1-on-1 counseling.